

Advanced (Carolina Reaper)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If two lines have slopes m_1 and m_2 , what is the condition for them to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 + m_2 = 0$
(E) $m_1 - m_2 = 1$
- Q3.** What is the equation of the line passing through $(1, 2)$ and perpendicular to $y = 3x - 1$?
(A) $y = -\frac{1}{3}x + \frac{7}{3}$
(B) $y = \frac{1}{3}x + \frac{5}{3}$
(C) $y = -3x + 5$
(D) $y = 3x - 1$
(E) $y = -\frac{1}{3}x + 2$
- Q4.** If a line has a slope of -2 , what is the slope of its perpendicular line?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q5.** Two lines have equations $y = 2x + 1$ and $y = -\frac{1}{2}x - 3$. What is their relationship?
(A) Parallel
(B) Perpendicular
(C) Skew
(D) Intersecting
(E) Coincident
- Q6.** What is the equation of a line perpendicular to $y = -4x + 2$ and passing through $(0, 0)$?
(A) $y = \frac{1}{4}x$
(B) $y = -\frac{1}{4}x$
(C) $y = 4x$
(D) $y = -4x$
(E) $y = 0$
- Q7.** If two lines have slopes -3 and $\frac{1}{3}$, what can be concluded about them?
(A) They are parallel
(B) They are perpendicular
(C) They are skew
(D) They are intersecting
(E) They are coincident
- Q8.** A line passes through $(2, 3)$ and is perpendicular to $y = x + 1$. What is its equation?
(A) $y = -x + 5$
(B) $y = x - 1$
(C) $y = -x + 3$
(D) $y = x + 2$
(E) $y = -x + 2$

Open Ended Questions (FRQ)

Q9. Prove that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x - 3$ are perpendicular. Show all work.

Q10. Find the equation of a line that passes through the point $(1, 4)$ and is perpendicular to the line $y = -2x + 1$. Show all work.